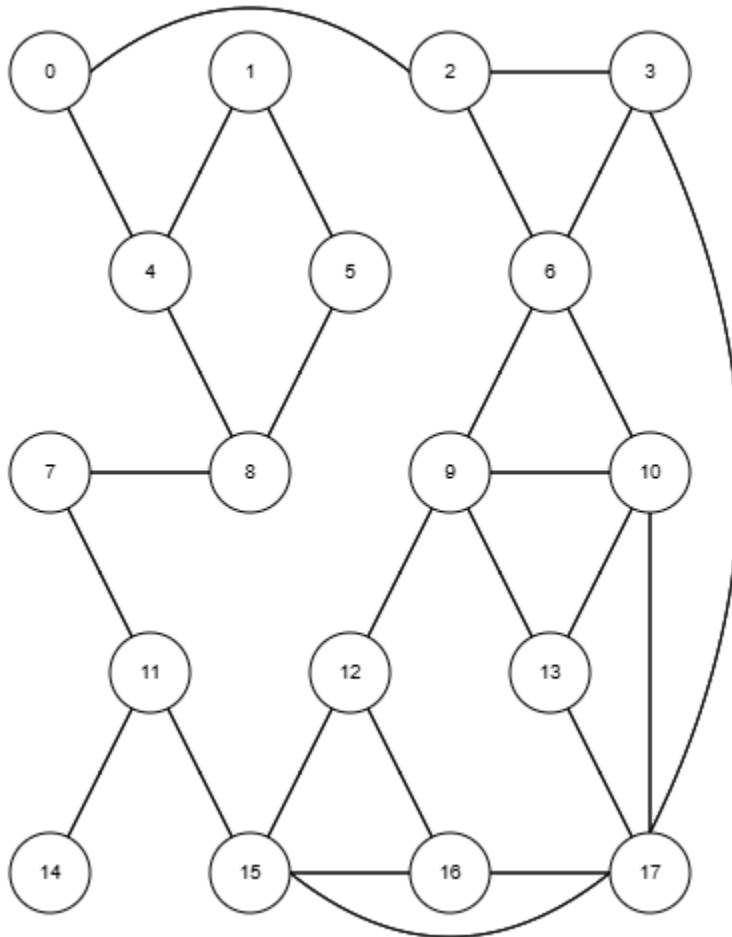


Activity

Q 1. Do DFS and BFS traversals starting with the vertex 1



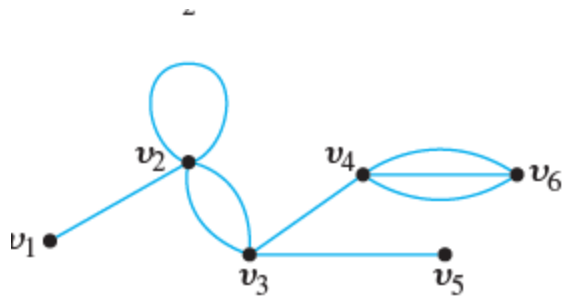
Q 2. A small social network contains three people who are network friends with six other people in the network, one person who is network friend with five other people in the network, and five people who are network friends with four other people in the network. The rest are network friends with three other people in the network. The network contains 41 pairs of network friends.

a. How many people are network friends with three other people in the network?

b. How many people are in the network?

Q 3. Graph with five vertices of degrees 1, 2, 3, 3, and 5.

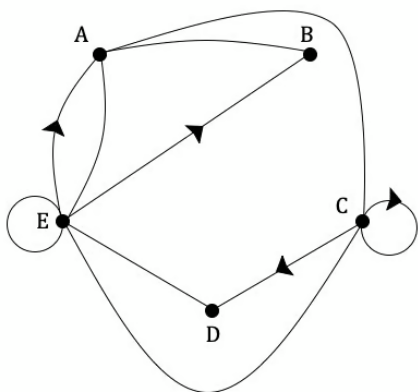
Q 4. Find the degree of each vertex and the total degree of the graph.



Q 5. Recall that $K_{m,n}$ denotes a complete bipartite graph on (m, n) vertices.

a. Draw $K_{4,2}$.

Q 6. Find the adjacency matrices for the following directed graph.



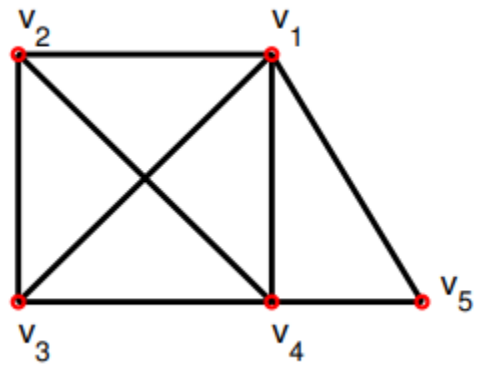
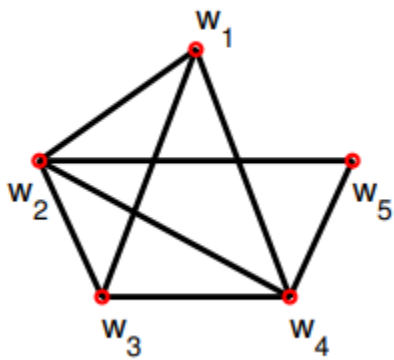
Q 7. The following is an adjacency matrix for a graph:

a. Draw the graph described by the adjacency matrix M .

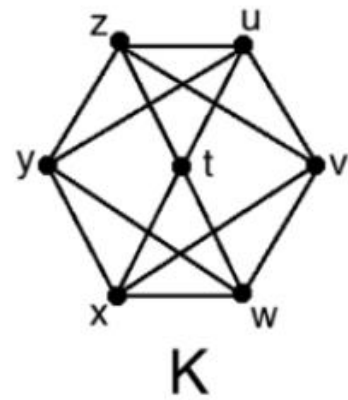
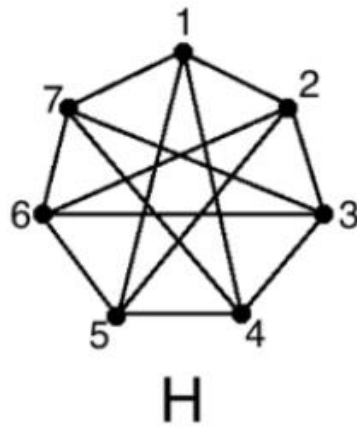
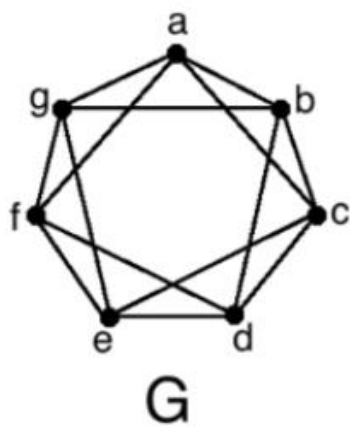
b. Find the number of walks of length 4 from vertex B to Vertex E.

$$M = \begin{matrix} & \begin{matrix} A & B & C & D & E \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Q 8. Determine whether the given pair of graphs is isomorphic. E



Q 9. Determine which pairs of graphs are isomorphic. If two graphs are isomorphic, show the isomorphism. If two graphs are not isomorphic



Q 10. Draw all nonisomorphic graphs with five vertices and no more than three edges.